

# 1. Project Introduction

## 1.1 Project Overview

The Restaurant Management System is a complete system designed to help restaurants manage their daily operations through one integrated platform instead of using manual work or paper records.

The system helps organize restaurant activities, reduce human errors, improve service speed, and make daily operations easier for both employees and management.

It covers the main restaurant operations such as employee management, customer management, menu management, table management, order processing, invoice generation, and business monitoring through a dashboard.

## 1.2 Project Idea

**The main idea of the project is to connect all restaurant departments into one system.**

**Instead of each department working separately, all information is connected automatically.**

**For example:**

- The cashier creates a new order.
- The selected table is automatically marked as Reserved.
- The order is saved in the system.
- After the customer finishes the meal, an invoice is generated.
- Finally, the table status changes back to Available.

This workflow reduces manual work, saves time, and improves the overall restaurant management process.

## 1.3 Project Scope

**The system manages the main operations inside the restaurant, including :**

- 1 - Employee Management
- 2 - Role Management
- 3 - Customer Management
- 4 - Restaurant Table Management
- 5 - Menu Management

- 6 - Category Management
- 7 - Order Management
- 8 - Order Details Management
- 9 - Invoice Generation
- 10 - Dashboard and Business Statistics

## **1.4 System Features**

**The Restaurant Management System provides many useful features, including :**

- 1 - Simple and user-friendly interface.
- 2 - Fast order creation using the POS screen.
- 3 - Automatic table status updates.
- 4 - PDF invoice generation.
- 5 - Business dashboard with important statistics.
- 6 - Search and sorting in all system pages.
- 7 - Clear validation and error messages.
- 8 - Role-based access control.
- 9 - Order status tracking (Pending, Preparing, Completed, Cancelled).
- 10 - Best-selling menu items statistics.
- 11 - Top employee performance statistics.
- 12 - Recent orders monitoring.
- 13 - Daily revenue tracking.
- 14 - Table availability monitoring.
- 15 - Complete restaurant operation management from one platform

## **1.5 Project Objectives**

**The main objectives of this project are :**

- 1 - Improve restaurant operation management.
- 2 - Reduce human errors.
- 3 - Increase service speed.
- 4 - Organize restaurant data. Improve customer service.
- 5 - Help managers monitor restaurant performance.
- 6 - Generate invoices automatically.
- 7 - Provide accurate business reports.
- 8 - Make decision-making easier using real-time statistics.
- 9 - Build a scalable system that can be extended with future features.

## **2. System Analysis**

### **2.1 Existing Problem**

Before developing the Restaurant Management System, many restaurant operations were performed manually or by using separate tools. This caused several operational problems and made restaurant management more difficult.

**Some of the common problems were:**

- 1 - Slow order registration.
- 2 - Human errors while recording customer orders.
- 3 - Difficulty tracking table availability.
- 4 - No clear monitoring of order status.
- 5 - Manual invoice preparation.
- 6 - Difficulty tracking daily revenue.
- 7 - No accurate statistics for best-selling menu items.
- 8 - Difficulty evaluating employee performance.
- 9 - Repeated customer information entry.
- 10 - Lack of centralized restaurant management.

These issues reduced work efficiency, increased operational errors, and affected customer satisfaction.

## 2.2 Proposed Solution

To solve these problems, the Restaurant Management System provides one integrated platform that manages all restaurant operations in one place.

The system allows each employee to perform their tasks according to their role while giving management complete visibility over restaurant activities.

**The system provides :**

- 1 - Employee Management
- 2 - Role Management
- 3 - Customer Management
- 4 - Restaurant Table Management
- 5 - Menu & Category Management
- 6 - Order Management
- 7 - POS (Point of Sale)
- 8 - Invoice Generation
- 9 - Dashboard with business statistics

This centralized solution improves workflow, reduces errors, saves time, and increases operational efficiency.

## 2.3 Stakeholders

The system is designed for different types of users. Each user has specific permissions according to their responsibilities.

### 1.Administrator

The Administrator has full access to the system

Responsibilities include:

- 1 - Managing employees
- 2 - Managing user roles
- 3 - Managing menu items and categories
- 4 - Managing customers
- 5 - Managing restaurant tables
- 6 - Viewing the dashboard
- 7 - Managing all restaurant data

## **2.Manager**

The Manager monitors the daily restaurant operations

Responsibilities include:

- 1 - Monitoring restaurant performance
- 2 - Tracking orders
- 3 - Viewing business statistics
- 4 - Monitoring employees
- 5 - Managing restaurant workflow

## **3.Cashier**

The Cashier handles customer orders and payments.

Responsibilities include:

- 1 - Creating new orders
- 2 - Using the POS screen
- 3 - Generating invoices
- 4 - Receiving customer payments

## **4.Waiter**

The Waiter assists customers inside the restaurant.

Responsibilities include:

- 1 - Serving customers
- 2 - Registering customer orders
- 3 - Monitoring table status

## **5.Chef**

The Chef is responsible for preparing customer orders.

Responsibilities include:

- 1 - Viewing incoming orders
- 2 - Tracking order status
- 3 - Preparing meals

## 2.4 Business Workflow

The Restaurant Management System follows the restaurant business process in the following sequence:

- 1 - Customer enters the restaurant.
- 2 - An available table is selected.
- 3 - A new order is created using the POS screen.
- 4 - Menu items and quantities are selected.
- 5 - The order is saved.
- 6 - The selected table becomes **Reserved** automatically.
- 7 - The kitchen prepares the order.
- 8 - The order is served to the customer.
- 9 - An invoice is generated.
- 10 - The order is completed.
- 11 - The table status automatically returns to **Available**.

This automated workflow improves restaurant efficiency and reduces manual work.

## 2.5 Business Benefits

Using the Restaurant Management System provides several business advantages:

- 1 - Faster order processing.
- 2 - Reduced human errors.
- 3 - Better customer service.
- 4 - Organized restaurant data.
- 5 - Accurate daily revenue tracking.
- 6 - Better employee performance monitoring.
- 7 - Automatic invoice generation.
- 8 - Easy access to restaurant information.
- 9 - Better business decision making through real-time statistics.

## **3. System Requirements**

### **3.1 Functional Requirements**

The Restaurant Management System provides several business functions that help manage restaurant operations efficiently.

#### **3.1.1 Employee Management**

**The system allows administrators to manage employee information.**

**Functions include:**

- 1 - Add new employees.
- 2 - Edit employee information.
- 3 - Delete employees.
- 4 - View employee details.
- 5 - Search employees.
- 6 - Sort employee records.
- 7 - Assign roles to employees.

#### **3.1.2 Role Management**

The system provides role management to control user permissions.

**Functions include:**

- 1 - Create new roles.
- 2 - Edit existing roles.
- 3 - Delete roles.
- 4 - View role details.
- 5 - Search roles.
- 6 - Prevent deleting roles assigned to employees

### **3.1.3 Customer Management**

The system stores customer information for future orders.

**Functions include:**

- 1 - Add new customers.
- 2 - Edit customer information.
- 3 - Delete customers.
- 4 - View customer details.
- 5 - Search customers.
- 6 - Reuse existing customer information while creating orders.

### **3.1.4 Category Management**

Categories help organize menu items.

**Functions include:**

- 1 - Add categories.
- 2 - Edit categories.
- 3 - Delete categories.
- 4 - Search categories.
- 5 - View category details.
- 6 - Prevent deleting categories that contain menu items.

### **3.1.5 Menu Management**

The system manages all food and drink items.

**Functions include:**

- 1 - Add menu items.
- 2 - Edit menu items.
- 3 - Delete menu items.
- 4 - Assign items to categories.
- 5 - Set item prices.
- 6 - Search menu items.
- 7 - View menu details.



### **3.1.6 Restaurant Table Management**

The system manages restaurant tables automatically.

**Functions include:**

- 1 - Add restaurant tables.
- 2 - Edit table information.
- 3 - Delete tables.
- 4 - View table details.
- 5 - Track table status.
- 6 - Automatically update table status after creating or completing orders.

### **3.1.7 Order Management**

The system manages customer orders from creation until completion.

**Functions include:**

- 1 - Create orders.
- 2 - Edit orders.
- 2 - Delete orders.
- 6 - View order details.
- 7 - Search orders.
- 7 - Sort orders.
- 8 - Track order status.
- 9 - Automatically calculate total amount
- 10 - Link order with customers , employees , tables

### **3.1.8 Order Details Management**

Each order contains one or more menu items.

**Functions include:**

- 1 - Add menu items to an order.
- 2 - Update quantities.
- 3 - Remove items.
- 4 - Calculate subtotal.
- 5 - Calculate order total automatically.

### **3.1.9 Point of Sale (POS)**

The POS screen is the main working screen for cashiers.

**Functions include:**

- 1 - Select customer.
- 2 - Select table.
- 3 - Add menu items.
- 4 - Update quantities.
- 5 - Remove menu items.
- 6 - Display order total instantly.
- 7 - Complete customer orders.
- 8 - Redirect to invoice after order completion.

### **3.1.10 Invoice Management**

The system generates invoices automatically.

**Functions include:**

- 1 - Display invoice details.
- 2 - Generate PDF invoices.
- 3 - Print invoices.
- 4 - Show customer information.
- 5 - Show ordered items.
- 6 - Show quantities and prices.
- 7 - Show total payment.

### **3.1.11 Dashboard**

The dashboard provides business statistics.

**It displays:**

- 1 - Total customers.
- 2 - Total employees.
- 3 - Total orders.
- 4 - Total revenue.

- 5 - Today's revenue.
- 6 - Today's orders.
- 7 - Available tables.
- 8 - Reserved tables.
- 9 - Pending orders.
- 10 - Preparing orders.
- 11 - Completed orders.
- 12 - Cancelled orders.
- 13 - Average order value.
- 14 - Top employee.
- 15 - Best-selling menu items.
- 16 - Recent orders.
- 17 - Orders during the last seven days

## **3.2 Non-Functional Requirements**

The Restaurant Management System should satisfy the following quality requirements.

### **Performance**

- 1 - The system should respond quickly.
- 2 - Orders should be processed in a short time.
- 3 - Dashboard statistics should load efficiently.

### **Usability**

- 1 - The interface should be simple.
- 2 - Users should learn the system easily.
- 3 - Navigation should be clear.

### **Security**

- 1 - User authentication is required.
- 2 - Users can only access authorized pages.
- 3 - Role-based permissions protect system data.

### **Reliability**

- 1 - The system should maintain data consistency.
- 2 - Business operations should be completed without data loss.

3 - Validation messages should prevent incorrect input.

### **Maintainability**

1 - The system should support future updates.

2 - New features can be added easily.

3 - Existing modules can be modified independently.

### **Availability**

1 - The system should be available whenever restaurant employees need it.

2 - Business operations should continue without interruption whenever possible.

## **4. System Design**

### **4.1 System Architecture**

The Restaurant Management System follows a layered architecture that separates the user interface, business logic, and database.

The system allows users to interact with the application through a simple interface. Every request is processed by the system, validated, and then stored or retrieved from the database.

#### **The architecture consists of the following components:**

1 - User Interface (UI)

2 - Business Logic

3 - Database

This architecture makes the system easier to maintain, organize, and extend in the future.

### **4.2 Database Design**

The database stores all restaurant information in a structured way.

The database was designed to reduce duplicated data and maintain relationships between different entities.

| Table            | Description                         |
|------------------|-------------------------------------|
| Employees        | Stores employee information         |
| Roles            | Stores user roles and permissions   |
| Customers        | Stores customer information         |
| RestaurantTables | Stores restaurant table information |
| Categories       | Stores menu categories              |
| MenuItems        | Stores food and drink items         |
| Orders           | Stores customers' orders            |
| OrderDetails     | Stored ordered menu items           |

### 4.3 Entity Relationship Diagram (ERD)

The Entity Relationship Diagram represents how all database tables are connected.

#### Main relationships include:

One Role can have many Employees.

One Customer can have many Orders.

One Employee can create many Orders.

One Restaurant Table can have many Orders.

One Category can contain many Menu Items.

One Order contains many Order Details.

One Menu Item can appear in many Order Detail

### 4.4 Main Business Entities

#### 1.Employee

Stores employee information such as:

- Name
- Job Title
- Salary
- Phone Number

- Username
- Password
- Role

## **2.Customer**

Stores customer information.

### **Information includes:**

- Customer Name
- Phone Number

## **3.Restaurant Table**

Represents restaurant tables.

### **Information includes:**

- Table Number
- Capacity
- Status (Available / Reserved)

## **4.Category**

Represents menu categories.

Examples:

- Pizza
- Burgers
- Drinks
- Desserts

## **5.Menu Item**

Represents food and drink items.

### **Information includes:**

- Item Name
- Description
- Price
- Category

## 6.Orders

**Represents customer orders.**

**Information includes:**

- Order Date
- Order Status
- Total Amount
- Customer
- Employee
- Restaurant Table

## 7.Order Details

Stores menu items inside each order.

**Information includes:**

- Menu Item
- Quantity
- Unit Price

## 4.5 Database Relationships

The database uses primary keys and foreign keys to connect tables together.

**Main relationships are:**

| Parent Table     | Child Table  | Relationship |
|------------------|--------------|--------------|
| Roles            | Employees    | One-to-Many  |
| Customers        | Orders       | One-to-Many  |
| Employees        | Orders       | One-to-Many  |
| RestaurantTables | Orders       | One-to-Many  |
| Categories       | MenuItems    | One-to-Many  |
| Orders           | OrderDetails | One-to-Many  |
| MenuItems        | OrderDetails | One-to-Many  |

## 4.6 Business Rules

The system follows several business rules to ensure correct restaurant operations.

### Examples include:

- 1 - Every employee must have one role.
- 2 - Every menu item must belong to one category.
- 3 - Every order must belong to one customer.
- 4 - Every order must be assigned to one employee.
- 5 - Every order must be assigned to one restaurant table.
- 6 - A table becomes Reserved after creating an order.
- 7 - A table becomes Available after completing the order.
- 8 - An invoice cannot be generated before an order exists.
- 9 - Order total is calculated automatically.
- 10 - Users can only access pages allowed by their assigned role.
- 11 - The system prevents deleting records that are linked to other important records.

## 4.7 System Workflow







